

the distribution of heart rates for the dogs when receiving Benzedrine is shifted to the right of that for the same animals when on the placebo. Use a one-tailed Wilcoxon signed-rank test with $\alpha = .05$.

Dog	Placebo	Benzedrine	Dog	Placebo	Benzedrine
1	250	258	8	296	305
2	271	285	9	301	319
3	243	245	10	298	308
4	252	250	11	310	320
5	266	268	12	286	293
6	272	278	13	306	305
7	293	280	14	309	313

6.6 Choosing Sample Sizes for Inferences about

$\mu_1 - \mu_2$

Sections 5.3 and 5.5 were devoted to sample-size calculations to obtain a confidence interval about μ with a fixed width and specified degree of confidence or to conduct a statistical test concerning μ with predefined levels for α and β . Similar calculations can be made for inferences about $\mu_1 - \mu_2$ with either independent samples or with paired data. Determining the sample size for a $100(1 - \alpha)\%$ confidence interval about $\mu_1 - \mu_2$ of width $2E$ based on independent samples is possible by solving the following expression for n . We will assume that both samples are of the same size.

$$z_{\alpha/2}\sigma\sqrt{\frac{1}{n} + \frac{1}{n}} = E$$

Note that, in this formula, σ is the common population standard deviation and that we have assumed equal sample sizes.

**Sample Sizes for a
100(1 - α)% Confidence
Interval for $\mu_1 - \mu_2$ of the
Form $\bar{y}_1 - \bar{y}_2 \pm E$,
Independent Samples**

$$n = \frac{2z_{\alpha/2}^2\sigma^2}{E^2}$$

(Note: If σ is unknown, substitute an estimated value to get an approximate sample size.)

The sample sizes obtained using this formula are usually approximate because we have to substitute an estimated value of σ , the common population standard deviation. This estimate will probably be based on an educated guess from information on a previous study or on the range of population values.

Corresponding sample sizes for one- and two-sided tests of $\mu_1 - \mu_2$ based on specified values of α and β , where we desire a level α test having the probability of a Type II error $\beta(\mu_1 - \mu_2) \leq \beta$ whenever $|\mu_1 - \mu_2| \geq \Delta$, are shown here.

Sample Sizes for Testing
 $\mu_1 - \mu_2$, Independent
Samples

$$\text{One-sided test: } n = 2\sigma^2 \frac{(z_\alpha + z_\beta)^2}{\Delta^2}$$

$$\text{Two-sided test: } n = 2\sigma^2 \frac{(z_{\alpha/2} + z_\beta)^2}{\Delta^2}$$

where $n_1 = n_2 = n$ and the probability of a Type II error is to be $\leq \beta$ when the true difference $|\mu_1 - \mu_2| \geq \Delta$. (Note: If σ is unknown, substitute an estimated value to obtain an approximate sample size.)

EXAMPLE 6.9

An experiment was done to determine the effect on dairy cattle of a diet supplemented with liquid whey. Whereas no differences were noted in milk production measurements among cattle given a standard diet (7.5 kg of grain plus hay by choice) with water and those on the standard diet and liquid whey only, a considerable difference between the groups was noted in the amount of hay ingested. Suppose that we test the null hypothesis of no difference in mean hay consumption for the two diet groups of dairy cattle. For a two-tailed test with $\alpha = .05$, determine the approximate number of dairy cattle that should be included in each group if we want $\beta \leq .10$ for $|\mu_1 - \mu_2| \geq .5$. Previous experimentation has shown σ to be approximately .8.

Solution From the description of the problem, we have $\alpha = .05$, $\beta \leq .10$ for $\Delta = |\mu_1 - \mu_2| \geq .5$, and $\sigma = .8$. Table 1 in the Appendix gives us $z_{.025} = 1.96$ and $z_{0.10} = 1.28$. Substituting into the formula, we have

$$n \approx \frac{2(.8)^2(1.96 + 1.28)^2}{(.5)^2} = 53.75, \text{ or } 54$$

Thus, we need 54 cattle per group to run the desired test.

Sample-size calculations can also be performed when the desired sample sizes are unequal, $n_1 \neq n_2$. Let n_2 be some multiple m of n_1 ; that is, $n_2 = mn_1$. For example, we may want n_1 three times as large as n_2 ; hence, $n_2 = \frac{1}{3}n_1$. The displayed formulas can still be used, but we must substitute $(m + 1)/m$ for 2 and n_1 for n in the sample-size formulas. After solving for n_1 , we have $n_2 = mn_1$.

EXAMPLE 6.10

Refer to Example 6.9. Suppose the experimenters wanted more information about the diet with liquid whey (group II) than about the diet with water (group I). In particular, the experimenters wanted 40% more cattle in group II than in group I; that is, $n_2 = (1.4)n_1$. All other specifications are as specified in Example 6.9.

Solution We replace 2 in our sample-size formula with $m = 1.4$. We then have

$$n \approx \frac{(m + 1)\sigma^2 (z_{\alpha/2} + z_\beta)^2}{m \Delta^2} = \frac{(1.4 + 1)(.8)^2 (1.96 + 1.28)^2}{1.4 (.5)^2} = 46.07, \text{ or } 47$$

That is, we need 47 cattle in group I and $(1.4)47 = 65.8$, or 66 in group II.

Sample sizes for estimating μ_d and conducting a statistical test for μ_d based on paired data (differences) are found using the formulas of Chapter 5 for μ . The only change is that we are working with a single sample of differences rather than a single sample of y values. For convenience, the appropriate formulas are shown here.

**Sample Size for a
100(1 - α)% Confidence
Interval for $\mu_1 - \mu_2$ of the
Form $\bar{d} \pm E$, Paired Samples**

$$n = \frac{z_{\alpha/2}^2 \sigma_d^2}{E^2}$$

(Note: If σ_d is unknown, substitute an estimated value to obtain approximate sample size.)

**Sample Sizes for Testing
 $\mu_1 - \mu_2$, Paired Samples**

$$\text{One-sided test: } n = \frac{\sigma_d^2 (z_\alpha + z_\beta)^2}{\Delta^2}$$

$$\text{Two-sided test: } n = \frac{\sigma_d^2 (z_{\alpha/2} + z_\beta)^2}{\Delta^2}$$

where the probability of a Type II error is β or less if the true difference $\mu_d \geq \Delta$. (Note: If σ_d is unknown, substitute an estimated value to obtain an approximate sample size.)

6.7 Summary

In this chapter, we have considered inferences about $\mu_1 - \mu_2$. The first set of methods was based on independent random samples being selected from the populations of interest. We learned how to sample data to run a statistical test or to construct a confidence interval for $\mu_1 - \mu_2$ using t methods. Wilcoxon's rank sum test, which does not require normality of the underlying populations, was presented as an alternative to the t test.

The second major set of procedures can be used to make comparisons between two populations when the sample measurements are paired. In this situation, we no longer have independent random samples, and hence the procedures of Sections 6.2 and 6.3 (t methods and Wilcoxon's rank sum) are inappropriate. The test and estimation methods for paired data are based on the sample differences for the paired measurements or the ranks of the differences. The paired t test and corresponding confidence interval based on the difference measurements were introduced and found to be identical to the single-sample t methods of Chapter 5. The nonparametric alternative to the paired t test is Wilcoxon's signed-rank test.

The material presented in Chapters 5 and 6 lays the foundation of statistical inference (estimation and testing) for the remainder of the text. Review the material in this chapter periodically as new topics are introduced so that you retain the basic elements of statistical inference.